

High Convolution Powers of Circle GMC Have Hölder Densities

Automated Math Discovery Candidate Partial Result

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Status. Candidate rigorous partial answer, likely valid pending expert review. The argument combines the later exact Fourier-dimension theorem for subcritical one-dimensional GMC with a deterministic Fourier-series lemma.

Source Problem

Garban and Vargas ask in [1]:

When are the convolutions of M_γ Hölder functions?

Here M_γ is subcritical Gaussian multiplicative chaos on the circle, $0 < \gamma < \sqrt{2}$. Their paper proves the Rajchman property and gives a nonoptimal Fourier-dimension lower bound for small γ .

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In conclusion, to summarize the above discussion, here is a list of open problems which seem interesting:

Open problems.

- Show the optimal asymptotic of c_n in the almost sure sense. (Theorem 1.3 only gives a strong hint of what this a.s. optimal asymptotic should be when $\gamma < \frac{1}{\sqrt{2}}$).
- When are the convolutions of M_γ Hölder functions?
- How can one prove Rajchman's property for the critical case $\gamma_c = \sqrt{2}$? (This question looks particularly interesting to us as our present technique would somehow require to consider an infinite convolution $M_{\gamma_c}^{*\infty}$ which is of course ill-defined).
- Is it the case that the Fourier dimension of M_γ is a.s. given by $s_*(\gamma)$? (Already for $\gamma < \frac{1}{\sqrt{2}}$, this question is open).

1.2.3 Relation to other models and results. To the best of our knowledge, Theorem (1.1) is the first proof of the Raichman property for a (random)

Lin, Qiu, and Tan subsequently proved the exact Fourier dimension predicted in the source paper [2]. We show that this immediately gives an explicit eventual-smoothing answer for every subcritical parameter.

Proof Intuition

Convolution turns multiplication of Fourier coefficients into ordinary powers. If a measure has Fourier coefficients $|\hat{\mu}(n)| \lesssim |n|^{-s/2}$, then its q -fold convolution has coefficients bounded by $|n|^{-qs/2}$.

Once $qs/2 > k + \alpha + 1$, the Fourier series remains absolutely summable after weighting by $|n|^{k+\alpha}$. This gives a $C^{k,\alpha}$ density. Taking s arbitrarily close to the exact Fourier dimension gives the claimed threshold.

Deterministic Smoothing Lemma

Write m for normalized Haar measure on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and use

$$\widehat{\mu}(n) = \int_{\mathbb{T}} e^{-in\theta} d\mu(\theta).$$

Proposition 1. *Let μ be a finite Borel measure on $\mathbb{T}$. Suppose that for some $s > 0$,*

$$|\widehat{\mu}(n)| \leq C(1 + |n|)^{-s/2} \quad (n \in \mathbb{Z}).$$

Let $q \geq 1$ and let $k \geq 0$ be integers. If $0 < \alpha < 1$ and

$$k + \alpha < \frac{qs}{2} - 1,$$

*then the q -fold convolution μ^{*q} has a $C^{k,\alpha}$ density with respect to m .*

Proof. The convolution identity gives

$$\widehat{\mu^{*q}}(n) = \widehat{\mu}(n)^q.$$

The assumed strict inequality implies

$$\sum_{n \in \mathbb{Z}} (1 + |n|)^{k+\alpha} |\widehat{\mu}(n)|^q \leq C^q \sum_{n \in \mathbb{Z}} (1 + |n|)^{k+\alpha-qs/2} < \infty.$$

Consequently the Fourier series

$$f_q(\theta) = \sum_{n \in \mathbb{Z}} \widehat{\mu}(n)^q e^{in\theta} \tag{1}$$

converges absolutely and uniformly. Its Fourier coefficients agree with those of μ^{*q} . Since trigonometric polynomials are dense in $C(\mathbb{T})$, uniqueness of Fourier coefficients for finite measures gives $d\mu^{*q} = f_q dm$.

The series obtained by differentiating (1) k times also converges uniformly. Moreover, using $|e^{inh} - 1| \leq 2|nh|^\alpha$, we have

$$|f_q^{(k)}(\theta + h) - f_q^{(k)}(\theta)| \leq 2|h|^\alpha \sum_{n \in \mathbb{Z}} |n|^{k+\alpha} |\widehat{\mu}(n)|^q.$$

Thus $f_q^{(k)}$ is α -Hölder, proving $f_q \in C^{k,\alpha}(\mathbb{T})$. □

Application to Gaussian Multiplicative Chaos

For $0 < \gamma < \sqrt{2}$, define

$$D_\gamma = \begin{cases} 1 - \gamma^2, & 0 < \gamma < 1/\sqrt{2}, \\ (\sqrt{2} - \gamma)^2, & 1/\sqrt{2} \leq \gamma < \sqrt{2}. \end{cases} \tag{2}$$

This is the correlation-dimension threshold denoted $s_*(\gamma)$ in the source paper.

Theorem 1. *Let M_γ be subcritical Gaussian multiplicative chaos on the circle. Almost surely, simultaneously for every integer $q \geq 1$, integer $k \geq 0$, and $0 < \alpha < 1$ satisfying*

$$k + \alpha < \frac{qD_\gamma}{2} - 1, \quad (3)$$

*the convolution M_γ^{*q} has a $C^{k,\alpha}$ density.*

In particular, if

$$qD_\gamma > 2,$$

*then M_γ^{*q} has a Hölder-continuous density of every exponent*

$$0 < \alpha < \frac{qD_\gamma}{2} - 1.$$

Thus a guaranteed first Hölder convolution power is

$$q_0(\gamma) = \left\lfloor \frac{2}{D_\gamma} \right\rfloor + 1.$$

Proof. The exact Fourier-dimension theorem of [2] states that, almost surely,

$$\dim_F(M_\gamma) = D_\gamma.$$

By the definition of Fourier dimension, for every $s < D_\gamma$ there is an almost surely finite random constant C_s such that

$$|\widehat{M}_\gamma(n)| \leq C_s(1 + |n|)^{-s/2} \quad (n \in \mathbb{Z}).$$

If (3) holds, choose $s < D_\gamma$ sufficiently close to D_γ that $k + \alpha < qs/2 - 1$. Proposition 1 then applies pathwise to M_γ . $\square$

Remark 1 (Scope). *The theorem proves eventual Hölder smoothing for every subcritical γ and gives all regularities forced by the known pointwise Fourier decay. It does not claim that the threshold $qD_\gamma > 2$ is necessary. Determining the first Hölder convolution power, including possible endpoint behavior, remains open.*

Novelty Check

Bounded searches through July 22, 2026 found the source question and the later exact Fourier-dimension papers, but no statement of the convolution corollary above. The proof is elementary once exact Fourier decay is known; novelty confidence is therefore moderate and should be checked by a specialist in GMC harmonic analysis.

References

- [1] C. Garban and V. Vargas, *Harmonic analysis of Gaussian multiplicative chaos on the circle*, arXiv:2311.04027 (2023).
- [2] Z. Lin, Y. Qiu, and M. Tan, *Harmonic analysis of multiplicative chaos, Part II: a unified approach to Fourier dimensions*, arXiv:2505.03298 (2025).